

Understanding Statistical Learning Theory (MA6056)

LECTURE SLIDES

July – November 2026

Topic 00 : Introduction and Information

- Linear Regression (A very quick review)
- Linear Model Selection and Regularization
- Classification
- Non-linear Regression
- Tree-based Methods
- Support Vector Machine
- Deep Learning
- Multiple Testing
- Survival Analysis

Text Books

- *An Introduction to Statistical Learning with Application in R* by G. James, D. Witten, T. Hastie, R. Tibshirani
- *An Introduction to Statistical Learning with Application in Python* by G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor

Reference Books

- *The Elements of Statistical Learning* by T. Hastie, R. Tibshirani, J. Friedman

Useful link

- <https://www.statlearning.com/>
- <https://hastie.su.domains/ElemStatLearn/>

Will be informed in the next week (Aug 04 – 06, 2026)

<https://ayonganguly.github.io/ma6056.html>

Instructor

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